

STUDENT SURVIVAL GUIDE: SAIGONRIDE FINAL PROJECT

Step-by-Step Instructions & Deliverable Checklist

This guide will walk you through the Software Development Life Cycle (SDLC) for your final project. You will work in groups of 2 or 3 to build **Saigon-Ride**, a distributed vehicle rental platform serving both Local Commuters and Foreign Tourists.

Directory of Visual Guides & Diagrams Included:

To help you succeed, this guide references the following architectural and conceptual diagrams (refer to your lecture slides for visual examples):

1. **Requirements Engineering Process:** Helps you extract and document FRs/NFRs.
2. **UML Use Case Diagram with Actor Generalization:** Shows how to model base users vs. specific users.
3. **CRC Cards:** Guides your object-oriented planning before drawing class diagrams.
4. **UI Design Progression:** Shows the 4 stages of Figma design you must submit.
5. **Git Workflow:** Illustrates how your team should collaborate on code.
6. **Boundary Value Analysis:** Explains the technique you must use for your QA unit tests.
7. **Requirements Traceability Matrix (RTM):** Shows how to map your final code to your initial requirements.

PART 1: Project Proposal & Requirements Engineering

(Due: Week 6-12Apr2026)

Total Value: 20% of Final Grade

What you need to do:

1. **Form Your Team & Methodology:** Decide roles for your 2-3 members. Choose your **SDLC methodology (e.g., Agile Scrum)** and justify why it works for a 7-week project.
2. **Assign 1 Core Use Case Per Student:** Your group must pick main functional workflows that include **complete CRUD** (Create, Read, Update, Delete) operations. Assign one specific Use Case to each student.

- *Example (Team of 3):* Student A owns "Manage Vehicles (CRUD)". Student B owns "Manage Stations (CRUD)". Student C owns "Process Rentals & Calculate Fares".
 - *Don't forget:* The team must also assign ownership of the **2 System Reports** (e.g., Revenue Report, Station Inventory Report).
3. **Identify the Actors:** Design the system for distinct users: The **Local Commuter**, the **Foreign Tourist**, and the **System Admin**.
 4. **Write Requirements (FRs & NFRs):** * Write Functional Requirements ("The system shall...") for your chosen use cases and reports.
 - Define at least 3 **Quality Attributes (NFRs)**. Make them measurable! (e.g., *Performance*: "Fare calculations must execute in under 1 second." *Security*: "Tourist passport data must be encrypted.")

Deliverable 1: Proposal & Requirements Document (PDF)

Your document must contain:

- Title Page (Project Name, Team Members, Tier Selection).
- Project Proposal (Roles, Methodology, Use Case ownership mapping).
- Problem Statement.
- Functional Requirements (FRs).
- Quality Attributes (NFRs).

Presentation 1: The Solution Pitch (5-7 mins)

- Prepare 3-5 slides. **Every member must speak** to explain the problem and present the specific Use Case they are taking ownership of.

PART 2: Software Design, Architecture & UI/UX

(Due: Week 20-26 Apr2026)

Total Value: 25% of Final Grade

What you need to do:

1. **Map User Interactions:** Draw an overall UML Use Case diagram showing Actor Generalization (Base User -> Local vs. Tourist).
2. **Deep Dive into Your Assigned Use Case:** For your individually assigned use case, write a formal **Use Case Specification** (Pre-conditions, Main Flow, Alternate Flows, Post-conditions) and draw a **Sequence Diagram** or **Activity Diagram**.

3. **Object & Quality Design:** * Create **CRC Cards** (Class-Responsibility-Collaborator) to plan how your objects interact before drawing your final classes.
 - Draw a formal UML Class Diagram or Entity-Relationship Diagram (ERD).
 - Draw your Architecture Diagram (3-Tier or MVC). Briefly write how your design uses **Quality Design principles** (like SOLID or Separation of Concerns) to meet the NFRs you defined in Part 1.
4. **Design the UI in Figma:** Create an interactive Prototype.
 - *Web Tiers:* Keep responsive **Bootstrap** grid layouts in mind!
 - *Features to show:* The 2 reports, the vehicle categories (Standard Bike vs. E-Scooter), and the checkout screen visually displaying buttons for Cash, Apple Pay, MoMo, VNPay, etc.

Deliverable 2: System Design Document (PDF)

Your document must contain:

- Overall Use Case Diagram.
- Detailed Specifications & Sequence/Activity diagrams for *each student's* Use Case.
- CRC Cards and Class Diagram/ERD.
- The Architectural Diagram + Quality Design explanation.
- **Figma Deliverable:** A public link to your Figma workspace.

Presentation 2: The Design Review (7-10 mins)

- Project your CRC cards/Class diagram and explain your database logic.
- Run your **Figma Prototype** live for the class.

PART 3 & FINAL SUBMISSION

(Due: Week 4-10 May2026)

Part 3 Value: 35% | Final Report & Video Value: 20%

What you need to do:

1. **Code Your Use Cases:** Implement the logic exactly as you designed it.
 - Ensure you code the dynamic pricing (Standard = 500 VND/min, Scooter = 1,500 VND/min) and the 15% location discount.
 - Use **Bootstrap** for styling if you choose a Web Tier.

- Both team members must push code to GitHub (>10 meaningful commits each).
- 2. **Detailed Design Justification:** Write a short section explaining what changed between your Part 2 UML diagrams and your actual Part 3 code. Did you have to add a new database table? Did the controller get split? Explain *why*.
- 3. **Perform QA Testing:** * Create a **Test Plan** table.
 - Write and execute Unit Tests targeting your pricing logic using **Boundary Value Analysis (BVA)** and **Equivalence Partitioning (EP)**.
 - *(Tier 4 Only: Write Automation Tests using Selenium/Playwright).*
- 4. **Verification (RTM):** Create a Requirements Traceability Matrix (RTM) to prove that your final code actually satisfies the FRs and NFRs you promised in Part 1.
- 5. **Compile the Final Report:** Merge your Part 1, 2, and 3 documents into a single PDF. Add **PDF bookmarks**. Add a **Lessons Learnt** section where each student reflects on their SDLC experience, challenges faced, and how they overcame them.
- 6. **Record a Video Demo:** Record your screen demonstrating the software working flawlessly.

Deliverables to Upload:

1. **Source Code:** A .zip of your entire project folder.
2. **Final Comprehensive PDF Report (10%):** Bookmarked, containing all SDLC phases, Detailed Design Justification, QA Test Plans, Verification (RTM), and Lessons Learnt.
3. **Video Demo Product (10%):** An .mp4 file or YouTube link explicitly demonstrating your chosen use cases and reports working end-to-end.

Presentation 3: Final Product Demo (10-15 mins)

- Live-demo the software to the class. Students will take turns presenting their specific use cases (with complete CRUD) working live. Open your GitHub history to prove everyone contributed to the codebase.